

2022.12

1. $\frac{n^3}{3}$ 相似题型 $A^2 B^2 C^4 X = b$ Gauss 消元法运算量 $\frac{2}{3}n^3$ ($A^2 B^2 C^4$ 共有 8 个矩阵)Poolittle 运算量 $\underline{n^3}$ ($\frac{1}{3} \cdot 3 = n^3$)

2. $\sum_{k=0}^{\infty} A^k = (I - A)^{-1}$

$\therefore$ 本题 $(I - a a^T)^{-1} = \begin{pmatrix} \frac{3}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{3}{2} \end{pmatrix}$

3. 均差性质: 设 $f(x)$ 在 $x_0, x_1, \dots, x_k$ 的区间 (a, b) 内 k 次可微, 则

$$f(x_0, x_1, \dots, x_k) = \frac{f^{(k)}(\xi)}{k!} = \frac{3 \cdot 3 \cdot 1}{3} = 3 \quad \text{for } f(0, 1, 2, 3) = \frac{3 \cdot 3 \cdot 1}{3} = 3$$

第 4 空: 若 $f(x) = x^m$, m 为自然数 则 $f(x_0, x_1, \dots, x_k) = \begin{cases} 0 & k > m \\ 1 & k = m \end{cases}$

$\therefore f(0, 2, 4, 6, 8) = 0$

4. $B_J = D^{-1}(L+U) \quad B_G = (D-L)^{-1}U$

$D = \begin{pmatrix} 2 & 1 & 2 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix} \quad L = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \end{pmatrix} \quad U = \begin{pmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$

$\det(\lambda I - B_J) = \begin{vmatrix} \lambda - \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ 1 & \lambda - \frac{1}{2} & 1 \\ -\frac{1}{2} & -\frac{1}{2} & \lambda - \frac{1}{2} \end{vmatrix} = \lambda(\lambda^2 + \frac{5}{4}) \quad \rho(B_J) > 1 \quad \text{发散}$

$\det(\lambda I - B_G) = \begin{vmatrix} 2\lambda & 1 & 1 \\ \lambda & \lambda & 1 \\ \lambda & \lambda & -2\lambda + 1 \end{vmatrix} = -\lambda(2\lambda + 1)^2 \quad \rho(B_G) < 1 \quad \text{收敛}$

5. $H(w) = I - \frac{2}{w^T w} w w^T \quad \therefore w^T w = 1 \quad w \text{ 为单位列向量}$

$x - H(w)x = \begin{pmatrix} \frac{1}{2} \\ \frac{3}{2} \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \\ \frac{3}{2} \end{pmatrix} \quad \text{单位化后 } w = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ 3 \end{pmatrix}$

6. $2x^3 + (1-a)x^2 + (b-5)x + (c-4) = 2(x+1)^3 \quad \therefore \underline{a=7 \quad b=11 \quad c=2}$

7. $x_{k+1} = \phi(x_k)$ 局部收敛 $x^* = 2 \quad \therefore |\phi'(2)| < 1 \quad \underline{0 < \beta < \frac{1}{2}}$

$\therefore$ 收敛到 $x^* = 2$ 则 $\phi'(2) = 0 \quad \phi''(2) \neq 0 \quad \therefore \underline{\beta = \frac{1}{4}}$

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$$8. x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)} = \frac{\sin x_k - x_k \cos x_k}{1 - \cos x_k}$$

$$9. [-1, 0] \quad [0, 1] \quad [2, 3]$$

$$10. \text{梯形求积公式: } \int_a^b f(x) dx = \frac{b-a}{6} (f(a) + 4f(\frac{a+b}{2}) + f(b))$$

梯形求积公式精度为3阶

$$11. \sum_{i=0}^n (x_i^4 - 12) |_{i=0} = (x^4 - 12)_{x=0} = -12$$

12. 梯形格式

$$u_{nH} = u_n + \frac{h}{2} (u_n' + u_{nH}')^2$$

$$\therefore \text{本空填 } u_{nH} = u_n + \frac{h}{2} (-u_n + t_n^2 - u_{nH} + t_{nH}^2)$$

$$= A^H A = \begin{pmatrix} 9 & 0 \\ 0 & 9 \end{pmatrix} \quad A^H A \text{ 的特征值 } \lambda_1 = 9 \quad \lambda_2 = 9 \quad \Sigma = \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix} \quad A^H A \text{ 对应的特征向量}$$

$$p_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, p_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad \text{标准正交化后 } V = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad U = AV \Sigma^{-1} = \begin{pmatrix} \frac{2}{3} & \frac{2}{3} \\ 0 & \frac{1}{3} \\ \frac{1}{3} & -\frac{2}{3} \end{pmatrix}$$

$$A = U \Sigma V^H = \begin{pmatrix} \frac{2}{3} & \frac{2}{3} \\ 0 & \frac{1}{3} \\ \frac{1}{3} & -\frac{2}{3} \end{pmatrix} \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$= L = \begin{pmatrix} 1 & & \\ l_{21} & 1 & \\ l_{31} & l_{32} & 1 \end{pmatrix} \quad U = \begin{pmatrix} u_{11} & u_{12} & u_{13} \\ & u_{22} & u_{23} \\ & & u_{33} \end{pmatrix}$$

$$u_{11} = 2 \quad u_{12} = 1 \quad u_{13} = 0$$

$$l_{21} = \frac{4}{2} = 2 \quad l_{31} = \frac{4}{4} = 1 \quad l_{22} = 5 - 2 = 3 \quad u_{23} = 1 \quad l_{32} = -1 \quad u_{33} = 3 + 1 = 4$$

$$A = LU = \begin{pmatrix} 1 & & \\ 2 & 1 & \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 2 & 1 & 0 \\ & 3 & 1 \\ & & 4 \end{pmatrix}$$

$$IV \quad \det(\lambda I - A) = (\lambda - 1)^2 (\lambda + 3) \quad \lambda_1 = \lambda_2 = 1 \quad \lambda_3 = -3$$

$$q(\lambda) = a\lambda^2 + b\lambda + c \quad f(\lambda) = e^\lambda$$

$$q(1) = a + b + c = e$$

$$q'(1) = 2a + b = e$$

$$q(-3) = 9a - 3b + c = e^3$$

求得 a. b. c

$$e^A = f(A) = aA^2 + bA + cI$$

本题用待定系数法计算复杂, 可用Jordan计算
(现已不要求掌握)

$$(2) V_1 = AV = (2, -2.999, 1)^T$$

$$\lambda_1 = -2.999$$

$$U_1 = (-0.66689, 1, 0.333)^T$$

$$\text{五. } a_0 = -\frac{1}{2} \quad a_1 = -\frac{1}{2} \quad a_2 = 1 \quad \beta_0 = \frac{\beta}{8} \quad \beta_1 = 1 \quad \beta_2 = \frac{4-\beta}{8}$$

$$C_0 = a_0 + a_1 + a_2 = 0$$

$$C_1 = a_1 + 2a_2 - (\beta_0\beta_1 + \beta_2) = 0$$

$$C_2 = \frac{1}{2}(a_1 + 2^2a_2) - (\beta_1 + 2\beta_2) = \frac{\beta-1}{4}, \quad \beta=1$$

$$C_3 = \frac{1}{6}(a_1 + 8a_2) - \frac{1}{2}(\beta_1 + 4\beta_2) = 0$$

$$C_4 = \frac{1}{24}(a_1 + 16a_2) - \frac{1}{6}(\beta_1 + 8\beta_2) = -\frac{1}{48} \neq 0$$

$\beta=1$ 时, 具有三阶精度 局部误差主项: $-\frac{1}{48}h^4u^{(4)}(t_n)$

$$(2) p(\lambda) = \lambda^2 - \frac{1}{2}\lambda - \frac{1}{2} = 0 \quad \lambda_1 = 1 \quad \lambda_2 = -\frac{1}{2} \quad \lambda_1 \text{ 是单根, } p=3 > 1 \therefore \text{收敛}$$

$$\text{四 } p(\lambda) - \bar{h}\sigma(\lambda) = 0$$

$$\lambda^2 - \frac{\frac{1}{2} + \bar{h}}{1 - \frac{3}{8}\bar{h}}\lambda - \frac{\frac{1}{2} + \frac{\bar{h}}{8}}{1 - \frac{3}{8}\bar{h}} = 0$$

$$\left| \frac{\frac{1}{2} + \bar{h}}{1 - \frac{3}{8}\bar{h}} \right| < 1 - \frac{\frac{1}{2} + \frac{\bar{h}}{8}}{1 - \frac{3}{8}\bar{h}} < 2$$

$$\therefore \bar{h} \in (-2, 0)$$

$$l_0 = \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)} = \frac{(x-x_1)(x-x_2)}{2} = \frac{x^2-x}{2}$$

$$l_1 = \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)} = 1-x^2$$

$$l_2 = \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)} = \frac{x^2+x}{2}$$

$$\therefore p(x) = \sum_{k=0}^2 y_k l_k = y_0 \frac{x^2-x}{2} + y_1 (1-x^2) + y_2 \frac{x^2+x}{2}$$

$$(2) f(x) = 1, x, x^2 \text{ 代入}$$

$$\int_{-1}^1 |x| dx = A_0 + A_1 + A_2$$

$$\int_{-1}^1 |x| x dx = A_0 x_0 + A_1 x_1 + A_2 x_2$$

$$\int_{-1}^1 |x| x^2 dx = A_0 x_0^2 + A_1 x_1^2 + A_2 x_2^2$$

$$A_0 = \frac{1}{4} \quad A_1 = \frac{1}{2} \quad A_2 = \frac{1}{4}$$

$$\text{当 } f(x) = x^3 \text{ 时 } \int_{-1}^1 |x| x^3 dx = 0 = A_0 x_0^3 + A_1 x_1^3 + A_2 x_2^3$$

$$\text{当 } f(x) = x^4 \text{ 时 } \int_{-1}^1 |x| x^4 dx \neq A_0 x_0^4 + A_1 x_1^4 + A_2 x_2^4$$

$\therefore$ 具有3次代数精度

$$t. \quad A B e^{tB} = B A e^{tA}$$

$$A \cdot e^{tB} B = B e^{tA} A$$

$$B e^{tA} B = A e^{tB} A$$

对 t 求导

$$B A e^{tA} B = A B e^{tB} A$$

$$B e^{tA} (AB) = A e^{tB} (BA)$$

$$AB = BA$$

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1. 把 $1, x, x^2, x^3$ 依次代入

$$\int_{-1}^1 (1-x^2) dx = A_1 + A_2 \quad (1)$$

$$\int_{-1}^1 (1-x^2)x dx = A_1 x_1 + A_2 x_2 \quad (2)$$

$$\int_{-1}^1 (1-x^2)x^2 dx = A_1 x_1^2 + A_2 x_2^2 \quad (3)$$

$$\int_{-1}^1 (1-x^2)x^3 dx = A_1 x_1^3 + A_2 x_2^3 \quad (4)$$

$$\text{由(2)得 } A_1 x_1 = -A_2 x_2$$

$$\text{代入(3)(4)} \quad A_1 x_1 (x_1 - x_2) = \frac{4}{15}$$

$$A_1 x_1 (x_1 - x_2) (x_1 + x_2) = 0$$

$$\text{得 } x_1 = -x_2 \quad \therefore A_1 = A_2 \quad \text{求得 } A_1 = A_2 = \frac{2}{3}$$

$$x_1 = \frac{1}{\sqrt{5}} \quad x_2 = -\frac{1}{\sqrt{5}}$$

$$2. \|A\|_{\infty} = 5 \quad \text{cond}_{\infty}(A) = \|A\|_{\infty} \|A^{-1}\|_{\infty} = 55$$

$$3. \text{代入得 } a_1 = 2 \quad a_3 = 4$$

$$4. \text{步长范围: } \begin{cases} u' = -10u \text{ 得 } u = 10 \\ u' = -u \end{cases} \quad 0 < h < \frac{2}{u} \quad \therefore \underline{0 < h < 0.2}$$

$$5. L = \begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix}$$

$$6. \sin A = \begin{pmatrix} \sin 0 & \cos 0 \\ & \sin 0 \\ & & \sin 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \sin 1 \end{pmatrix} \quad e^{At} = \begin{pmatrix} e^{0 \cdot t} & t e^{0 \cdot t} & 0 \\ 0 & e^{0 \cdot t} & 0 \\ 0 & 0 & e^{1 \cdot t} \end{pmatrix} = \begin{pmatrix} 1 & t & 0 \\ 0 & 1 & 0 \\ 0 & 0 & e^t \end{pmatrix}$$

$$7. x^3 + ax^2 + 27x + 27 = (x+3)^3$$

$$-2x^3 + (3-a)x^2 + (b-25)x - 2 = -2(x+1)^3$$

$$x^3 + (c-3)x^2 + (19-b)x = x^3$$

$$a=9 \quad b=19 \quad c=3$$

$$8. f(0,1,2,3) = \frac{f'''(3)}{3!} = \frac{2 \cdot 3!}{3!} = 2 \quad f(0,1,2,3,4) = 0$$

9. 高次 Lagrange 插值不一定收敛, 有时会出现 Runge/龙格现象

$$10. \underline{p(b)} < 1 \quad \underline{\|B\|} < 1$$

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$$A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 3 & 1 & 2 \\ 1 & 0 & 3 \end{pmatrix} = LU$$

$$AX=b \Rightarrow \begin{cases} Ly=b \\ UX=y \end{cases} \Rightarrow y = \begin{pmatrix} 0 \\ 0 \\ 3 \end{pmatrix}, x = \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}$$

$$\text{三. } \begin{pmatrix} \sum_{i=0}^3 x_i & \sum_{i=0}^3 x_i^2 & \sum_{i=0}^3 x_i^3 \\ \sum_{i=0}^3 x_i^2 & \sum_{i=0}^3 x_i^3 & \sum_{i=0}^3 x_i^4 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} \sum_{i=0}^3 y_i \\ \sum_{i=0}^3 x_i y_i \\ \sum_{i=0}^3 x_i^2 y_i \end{pmatrix} \quad y = a + bx + cx^2$$

$$\begin{pmatrix} 4 & 2 & 6 \\ 2 & 6 & 8 \\ 6 & 8 & 18 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 1 \\ -5 \\ -1 \end{pmatrix} \quad a=0.45 \quad b=-0.65 \quad c=0.25$$

$$\therefore y = 0.45 - 0.65x - 0.25x^2$$

$$\text{四. } f(x) = x^3 - a$$

$$x_{k+1} = x_k - \frac{x_k^3 - a}{3x_k^2} = \varphi(x_k)$$

$$\varphi'(a) = 0 \quad \varphi''(a) \neq 0 \quad \therefore \text{二阶收敛}$$

五. (不考)

六. Jacobi 迭代格式

$$\begin{aligned} x_1^{(k+1)} &= \frac{1}{3}(1 - x_2^{(k)} + x_3^{(k)}) \\ x_2^{(k+1)} &= -\frac{1}{5}(1 - 2x_1^{(k)} - 2x_3^{(k)}) \\ x_3^{(k+1)} &= \frac{1}{8}(1 - x_1^{(k)} - 6x_2^{(k)}) \end{aligned} \quad x^{(0)} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \quad x^{(1)} = \begin{pmatrix} \frac{1}{3} \\ -\frac{1}{5} \\ \frac{1}{8} \end{pmatrix} \quad x^{(2)} = \begin{pmatrix} \frac{53}{120} \\ -\frac{1}{60} \\ \frac{7}{30} \end{pmatrix}$$

Gauss-Seidel 迭代格式

$$\begin{aligned} x_1^{(k+1)} &= \frac{1}{3}(1 - x_2^{(k)} + x_3^{(k)}) \\ x_2^{(k+1)} &= -\frac{1}{5}(1 - 2x_1^{(k+1)} - 2x_3^{(k)}) \\ x_3^{(k+1)} &= \frac{1}{8}(1 - x_1^{(k+1)} - 6x_2^{(k+1)}) \end{aligned} \quad x^{(1)} = \begin{pmatrix} \frac{1}{3} \\ -\frac{1}{15} \\ \frac{2}{15} \end{pmatrix} \quad x^{(2)} = \begin{pmatrix} \frac{4}{15} \\ -\frac{3}{75} \\ \frac{73}{600} \end{pmatrix}$$

判断是否收敛:

$$L = \begin{pmatrix} 0 & 0 & 0 \\ -1 & 0 & 0 \\ -1 & -6 & 0 \end{pmatrix} \quad U = \begin{pmatrix} 0 & 1 & 1 \\ 0 & 0 & -2 \\ 0 & 0 & 0 \end{pmatrix}$$

$$B_J = D^{-1}(L+U) = \begin{pmatrix} 0 & -\frac{1}{3} & \frac{1}{3} \\ \frac{2}{3} & 0 & \frac{2}{3} \\ -\frac{1}{3} & -\frac{2}{3} & 0 \end{pmatrix}$$

$$\|B_J\|_\infty = \frac{7}{3} < 1 \quad \therefore \text{Jacobi 迭代收敛}$$

$$B_G = (D-L)^{-1}U$$

$$\det(\lambda I - B_G) = \begin{vmatrix} 3\lambda & 1 & -1 \\ 2\lambda & -5\lambda & 2 \\ \lambda & 6\lambda & 4\lambda \end{vmatrix} = \lambda(-120\lambda^2 - 69\lambda + 2) \quad B_G \text{ 的最大特征值小于 } 1$$

$$\therefore \rho(B_G) < 1 \quad \therefore \text{Gauss-Seidel 迭代收敛}$$

$$t. \quad A^H A = \begin{pmatrix} 9 & 0 \\ 0 & 1 \end{pmatrix} \quad \Sigma = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix} \quad A^H A \text{ 的特征值 } \lambda_1 = 9 \quad \lambda_2 = 1$$

$$\text{对应的特征向量 } p_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad p_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad v = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$U = AV\Sigma^{-1} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \\ 0 & 0 \end{pmatrix}$$

$$A = U\Sigma V^H = \begin{pmatrix} 0 & 1 \\ 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

2021.6

A的特征值为4,0 A为非零正规阵 $\|A\|_2 = \sqrt{\lambda_{\max}(A^T A)} = \lambda_{\max} A = 4$

$H(W) = I - \frac{2}{W^T W} W W^T$ $\therefore W^T W = 1$ W为单位列向量

$$X - H(W)X = \begin{pmatrix} 2 \\ -1 \\ -2 \end{pmatrix} - \begin{pmatrix} 3 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \\ -2 \end{pmatrix} \quad \text{单位化后 } W = \begin{pmatrix} -\frac{1}{\sqrt{6}} \\ -\frac{1}{\sqrt{6}} \\ -\frac{2}{\sqrt{6}} \end{pmatrix}$$

$X_{k+1} = P(X_k)$ 局部收敛到 $X^* = \sqrt{6}$ $\therefore |P'(\sqrt{6})| < 1 \Rightarrow 0 < \alpha < \sqrt{6}$

$$X_{k+1} = X_k - \frac{f(X_k)}{f'(X_k)} = \frac{\sin X_k - X_k \cos X_k}{1 - \cos X_k}$$

$$A^T = \begin{pmatrix} 5 & 4 \\ 21 & 21 \\ -21 & 21 \end{pmatrix} \quad V^{(6)} = A^T U^{(5)} = \begin{pmatrix} -0.3334 \\ -0.1667 \end{pmatrix}, \quad \lambda_2 = \frac{1}{|V^{(6)}|_{\max}} = \frac{1}{0.3334} = 2.9994$$

$$X^{(2)} = \frac{V^{(6)}}{\lambda_2} = \begin{pmatrix} 1.0000 \\ -0.5000 \end{pmatrix}$$

$$5. \int_{-1}^1 f(x) dx \approx \sum_{i=0}^4 A_i f(x_i)$$

$$\sum_{i=0}^4 A_i (4x_i^3 - 3x_i^2) = \int_{-1}^1 (4x^3 - 3x^2) dx = -2$$

$$\sum_{i=0}^4 A_i (x_i^6 - 6x_i^5) = \int_{-1}^1 (x^6 - 6x^5) dx = \frac{2}{7}$$

$$6. \text{梯形格式 } U_{n+1} = U_n + \frac{h}{2} (u'_n + u'_{n+1})$$

$$\therefore U_{n+1} = U_n + \frac{h}{2} (-u_n^2 + t_n + t_{n+1} - u_{n+1}^2)$$

$$7. \text{Euler} \quad -\beta \tau_1 \quad \frac{1}{2} h^2 u''(t)$$

$$\text{改进 Euler} \quad -\beta \tau_1 \quad -\frac{1}{2} h^2 u''(t)$$

$$\text{梯形格式} \quad -\beta \tau_1 \quad -\frac{1}{12} h^3 u'''(t)$$

$$8. a_0 = -\frac{1}{2} \quad a_1 = -\frac{1}{2} \quad a_2 = 1 \quad \beta_0 = \frac{1}{8} \quad \beta_1 = 1 \quad \beta_2 = \frac{3}{8}$$

$$p(\lambda) = \lambda^2 - \frac{1}{2}\lambda - \frac{1}{2} \quad \sigma(\lambda) = \frac{3}{8}\lambda^2\lambda + \frac{1}{8} \quad p(\lambda) - \bar{h}\sigma(\lambda) = 0$$

$$\lambda^2 - \frac{\frac{1}{2} + \bar{h}}{1 - \frac{3}{8}\bar{h}} \lambda - \frac{\frac{1}{2} + \frac{1}{8}\bar{h}}{1 - \frac{3}{8}\bar{h}} = 0$$

$$\left| \frac{\frac{1}{2} + \bar{h}}{1 - \frac{3}{8}\bar{h}} \right| < 1 - \frac{\frac{1}{2} + \frac{1}{8}\bar{h}}{1 - \frac{3}{8}\bar{h}} < 2$$

$$\bar{h} \in (-2, 0)$$

$$9. p(x) = (4x-3)(x+2)x - 1$$

$$=, u_m = \int_a^b p(x) x^m dx$$

$$u_0 = \int_0^1 x dx = \frac{1}{2} \quad u_1 = \int_0^1 x^2 dx = \frac{1}{3} \quad u_2 = \int_0^1 x^3 dx = \frac{1}{4} \quad u_3 = \int_0^1 x^4 dx = \frac{1}{5}$$

$$\phi_0(x) = 1 \quad \phi_1(x) = \begin{vmatrix} u_0 & 1 \\ u_1 & x \end{vmatrix} = \frac{1}{2}x - \frac{1}{3}$$

$$\phi_2(x) = \begin{vmatrix} u_0 & u_1 & 1 \\ u_1 & u_2 & x \\ u_2 & u_3 & x^2 \end{vmatrix} = \frac{1}{72}x^2 - \frac{1}{60}x + \frac{1}{240}$$

$$(2) p'(0) = p'(1) = 0, \quad x=0, x=1 \text{ 有二重根}$$

$$p[0,0] = p'(0)$$

$$p[1,1] = p'(1)$$

x	$p(x)$	一阶均差	二阶均差	三阶均差
0	2			
0	2	0		
1	0	-2	2	
1	0	0		4

$$p(x) = 2 - 2x^2 + 4x^2(x-1) = 4x^3 - 6x^2 + 2$$

三、将 $f(x) = 1, x, x^2, x^3$ 代入

$$\int_{-1}^1 1 dx = A_0 + A_1 + A_2$$

$$\int_{-1}^1 x dx = A_0 x_0 + A_2 x_2$$

$$\int_{-1}^1 x^2 dx = A_0 x_0^2 + A_2 x_2^2$$

$$\int_{-1}^1 x^3 dx = A_0 x_0^3 + A_2 x_2^3$$

$$\int_{-1}^1 x^4 dx = A_0 x_0^4 + A_2 x_2^4$$

$$\int_{-1}^1 x^5 dx = A_0 x_0^5 + A_2 x_2^5$$

$$A_0 = \frac{5}{9} \quad A_1 = \frac{8}{9} \quad A_2 = \frac{5}{9}$$

$$\int_{-1}^1 x^6 dx \neq A_0 x_0^6 + A_2 x_2^6$$

代数精度为 5

$\Rightarrow$ (解方程组可参看 2021.12 试卷第 1 题)

四. $y = a + bx$

$$\begin{pmatrix} \sum_{i=0}^4 1 & \sum_{i=0}^4 x_i \\ \sum_{i=0}^4 x_i & \sum_{i=0}^4 x_i^2 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} \sum_{i=0}^4 y_i \\ \sum_{i=0}^4 x_i y_i \end{pmatrix}$$

$$\begin{pmatrix} 5 & 15 \\ 15 & 55 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 50 \\ 200 \end{pmatrix}$$

$$a = -5 \quad b = 5$$

$$y = -5 + 5x$$

五. $A = LU$

$$L = \begin{pmatrix} 1 & & \\ l_{21} & 1 & \\ l_{31} & l_{32} & 1 \end{pmatrix} \quad U = \begin{pmatrix} u_{11} & u_{12} & u_{13} \\ & u_{22} & u_{23} \\ & & u_{33} \end{pmatrix}$$

$$u_{11} = 1 \quad u_{12} = 1 \quad u_{13} = 0$$

$$l_{21} = 2 \quad u_{22} = 2 \quad u_{23} = 3$$

$$l_{31} = 2 \quad l_{32} = -1 \quad u_{33} = 8$$

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 2 & 1 & 3 \\ 2 & -1 & 8 \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 2 & 3 \\ 0 & 0 & 8 \end{pmatrix}$$

(2) $x_1^{(k+1)} = -2 - x_2^{(k)}$

$$x_2^{(k+1)} = \frac{1}{4} (3 - 2x_1^{(k)} - 3x_3^{(k)})$$

$$x_3^{(k+1)} = \frac{1}{5} (3 - 2x_1^{(k+1)})$$

$$x^0 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$x^{(1)} = \begin{pmatrix} 3 \\ 0 \\ \frac{9}{5} \end{pmatrix}$$

$$x^{(2)} = \begin{pmatrix} -2 \\ -\frac{11}{10} \\ \frac{7}{5} \end{pmatrix}$$

(3) $\det(\lambda I - B_4) = \begin{vmatrix} \lambda & 1 & 0 \\ 2\lambda & 4\lambda & 3 \\ 2\lambda & 0 & 5\lambda \end{vmatrix} = \lambda(20\lambda^2 - 10\lambda + 6)$

$\rho(B_4) = \lambda_{\max} B_4 < 1$ $\therefore$ Gauss-seidel 迭代收敛

六. $q(\lambda) = a\lambda + b \quad f(\lambda) = e^{\lambda t}$

$$q(2) = 2a + b = e^{2t}$$

$$q'(2) = a = te^{2t}$$

$$a = te^{2t} \quad b = (1-2t)e^{2t}$$

$$q(\lambda) = te^{2t}\lambda + (1-2t)e^{2t}$$

$$e^{At} = te^{2t}A + (1-2t)e^{2t}I$$

$$e^{At} = \begin{pmatrix} 1 & 0 \\ 3te^{2t} & 1 \end{pmatrix}$$

(2) $A^H A = \begin{pmatrix} 13 & 6 \\ 6 & 4 \end{pmatrix}$ $\det(\lambda I - A^H A) = (\lambda - 1)(\lambda - 16)$ $A^H A$ 的特征值 $\lambda_1 = 1$ $\lambda_2 = 16$
 $\Sigma = \begin{pmatrix} 1 & 0 \\ 0 & 4 \end{pmatrix}$ $A^H A$ 的对应特征向量 $P_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ $P_2 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ 正交化后 $V = \begin{pmatrix} \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \\ -\frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \end{pmatrix}$

$$U = AV\Sigma^{-1} = \begin{pmatrix} \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \\ -\frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \end{pmatrix}$$

$$A = U\Sigma V^H = \begin{pmatrix} \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \\ -\frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 4 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{5}} & -\frac{2}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \end{pmatrix}$$

2. $a_0 = 1$ $a_1 = a_2 = 0$ $a_3 = 1$ $\beta_0 = 0$ $\beta_1 = \frac{9}{4}$ $\beta_2 = 0$ $\beta_3 = \frac{3}{4}$

$$L_0 = a_0 + a_1 + a_2 + a_3 = 0$$

$$C_1 = a_1 + 2a_2 + 3a_3 - (\beta_0 + \beta_1 + \beta_2 + \beta_3) = 0$$

$$C_2 = \frac{1}{2}(a_1 + 4a_2 + 9a_3) - (\beta_1 + 2\beta_2 + 3\beta_3) = 0$$

$$C_3 = \frac{1}{6}(a_1 + 8a_2 + 27a_3) - \frac{1}{2}(\beta_1 + 4\beta_2 + 9\beta_3) = 0$$

$$C_4 = \frac{1}{24}(a_1 + 2^4 a_2 + 3^4 a_3) - \frac{1}{6}(\beta_1 + 8\beta_2 + 27\beta_3) = -\frac{9}{24}$$

$\therefore$ 局部截断误差主项: $-\frac{9}{24} h^4 u^{(4)}(t_n)$

$$p(\lambda) = \lambda^3 - 1 = 0 \quad \lambda_1 = \lambda_2 = \lambda_3 = 1 \quad \lambda = 1 \text{ 不满足单根条件}$$

$\therefore$ 发散,

2018.6

$$1. \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = R$$

$$\therefore \text{收敛半径 } R = \lim_{k \rightarrow \infty} \frac{|k+1|^4}{k^4} = 1 \quad \because \|A\| < 1 \Rightarrow \text{绝对收敛}$$

$$2. f[0, 1, 2, 3] = \frac{f'''(3)}{3!} = \frac{2 \cdot 3!}{3!} = 2 \quad f[0, 1, 2, 3, 4] = 0$$

$$3. \frac{|x-a|}{|a|} \leq \frac{1}{2a_1} \times 10^{1-n}$$

$$\sqrt{70} = 8.366600265 \dots \quad a_1 = 8$$

$$\frac{|\sqrt{70}-a|}{|a|} \leq \frac{1}{2 \times 8} \times 10^{1-n} \leq 0.1\%$$

$\therefore n=3$ 3位有效数字

$$\text{则 } a = 8.37$$

4. 梯形求积公式代数精度为三阶

$$5. L = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{pmatrix} \quad \|A\|_1 = 17 \quad \|AX\|_\infty = 23$$

$$6. W = \begin{pmatrix} -\frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{pmatrix} \quad \|W\|_2 = 1$$

$$7. v^{(8)} = A u^{(7)} = \begin{pmatrix} 3.499997 \\ 6.999996 \end{pmatrix} \quad \lambda = \|v^{(8)}\|_{\max} = 6.999996$$

$$8. \begin{pmatrix} \sum_{i=0}^4 1 & \sum_{i=0}^4 x_i \\ \sum_{i=0}^4 x_i & \sum_{i=0}^4 x_i^2 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} \sum_{i=0}^4 y_i \\ \sum_{i=0}^4 x_i y_i \end{pmatrix} \quad a = \frac{3}{5} \quad b = \frac{11}{10} \quad y = \frac{3}{5} + \frac{11}{10}x$$

$$\text{法二: 求解 } u = -\frac{10}{11}x$$

9. $x_{k+1} = \varphi(x_k)$ 局部平方收敛 $x^* = \sqrt{15}$

10. $\varphi(\sqrt{15}) = \varphi'(\sqrt{15}) = 0$ $\varphi''(\sqrt{15}) \neq 0$ $\delta = \frac{1}{2\sqrt{15}}$

11. $x_{k+1} = x_k - \frac{e^{x_k} + e^{-x_k} - 2}{e^{x_k} - e^{-x_k}}$

12. $x^3 + ax^2 - 3x^2 = (x-0)^3 \quad \therefore a=3$

13. A 为正规阵

14. $\int_{-1}^1 f(x) dx = \sum_{k=0}^4 A_k f(x_k) \quad \sum_{k=0}^4 A_k (x_k^2 - 2) = \int_{-1}^1 (x^2 - 2) dx = -\frac{10}{3}$

$\sum_{k=0}^4 A_k (5x_k^6 - 3x_k^3) = \int_{-1}^1 (5x^6 - 3x^3) dx = \frac{10}{7}$

15. $\sum_{i=0}^n x_i^2 li(0) = (x)_{x=0}^2 = 0 \quad \sum_{i=0}^n (x_i^4 + 2) li(x) = x^4 + 12$

16. $u_{n+1} = u_n + h(u'_{n+1}) \Rightarrow$ 隐 Euler 格式

$\therefore$ 本空填 $u_{n+1} = u_n + h(-u_{n+1} + t_{n+1})$

隐 Euler 是 -1 阶

Euler 是 -1 阶

梯形是 -2 阶

16. LU 分解

$$\therefore A = LL^T$$

$$= \begin{pmatrix} L_{11} & L_{12} & L_{13} \\ L_{21} & L_{22} & L_{23} \\ L_{31} & L_{32} & L_{33} \end{pmatrix} \begin{pmatrix} L_{11} & L_{12} & L_{13} \\ L_{22} & L_{23} & L_{33} \end{pmatrix}$$

$$L = \begin{pmatrix} 2 & & \\ \frac{1}{2} & 2 & \\ 0 & \frac{1}{2} & 2 \end{pmatrix} \quad A = \begin{pmatrix} 2 & \frac{1}{2} & 0 \\ \frac{1}{2} & 2 & 2 \\ 0 & \frac{1}{2} & 2 \end{pmatrix}$$

$$\begin{aligned} (2) \quad x_1^{(k+1)} &= \frac{1}{4} (8 - x_2^{(k)}) \\ x_2^{(k+1)} &= \frac{1}{4.25} (22 - x_1^{(k+1)} - x_3^{(k)}) \\ x_3^{(k+1)} &= \frac{1}{4.25} (38 - x_2^{(k+1)}) \end{aligned}$$

$$(3) \quad P_4 = (D - L)^{-1} U$$

$$\det(\lambda I - P_4) = \begin{vmatrix} 4\lambda & 1 & 0 \\ \lambda & 4.25\lambda & 1 \\ 0 & \lambda & 4.25\lambda \end{vmatrix} = \lambda^2 \left(\lambda - \frac{33}{17} \right)$$

$$\rho(P) < 1 \quad \therefore \text{收敛}$$

三、待定系数法 计算复杂 可用 Jordan 计算

$$\text{四. } u_m = \int_0^1 p(x) x^m dx \quad u_0 = \int_0^1 x dx = \frac{1}{2} \quad u_1 = \int_0^1 x^2 dx = \frac{1}{3} \quad u_2 = \int_0^1 x^3 dx = \frac{1}{4}$$

$$u_3 = \int_0^1 x^4 dx = \frac{1}{5} \quad \phi_0(x) = 1 \quad \phi_1(x) = \begin{vmatrix} u_0 & 1 \\ u_1 & x \end{vmatrix} = \frac{1}{2}x - \frac{1}{3}$$

$$\phi_2(x) = \begin{vmatrix} u_0 & u_1 & 1 \\ u_1 & u_2 & x \\ u_2 & u_3 & x^2 \end{vmatrix} = \frac{1}{72}x^2 - \frac{1}{60}x + \frac{1}{240}$$

$$(2) \quad f(x) = 1, x, x^2, x^3 \text{ 代 } \lambda$$

$$u_0 = \int_0^1 x dx = A_0 + A_1$$

$$u_1 = \int_0^1 x^2 dx = A_0 x_0 + A_1 x_1$$

$$u_2 = \int_0^1 x^3 dx = A_0 x_0^2 + A_1 x_1^2$$

$$u_3 = \int_0^1 x^4 dx = A_0 x_0^3 + A_1 x_1^3$$

$$五. f(x) = \frac{1}{2}(x^T A^T - b^T)(Ax - b)$$

$$= \frac{1}{2}(x^T A^T A x - x^T A^T b - b^T A x + b^T b)$$

$$\frac{df(x)}{dx} = 2A^T A x - A^T b - (b^T A)^T = 2(A^T A x - A^T b) = 0$$

$$A^T A x = A^T b$$

$$(2) A^T A = \begin{pmatrix} 9 & 0 \\ 0 & 5 \end{pmatrix} \quad A^T A \text{ 的特征值 } \lambda_1 = 9 \quad \lambda_2 = 5 \text{ 对应的特征向量 } p_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad p_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad V = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\Sigma = \begin{pmatrix} 3 & 0 \\ 0 & \sqrt{5} \end{pmatrix} \quad U = AV\Sigma^{-1} = \begin{pmatrix} \frac{2}{3} & \frac{1}{\sqrt{5}} \\ -\frac{1}{3} & \frac{2}{\sqrt{5}} \\ \frac{2}{3} & 0 \end{pmatrix}$$

$$A = U\Sigma V^T = \begin{pmatrix} \frac{2}{3} & \frac{1}{\sqrt{5}} \\ -\frac{1}{3} & \frac{2}{\sqrt{5}} \\ \frac{2}{3} & 0 \end{pmatrix} \begin{pmatrix} 3 & 0 \\ 0 & \sqrt{5} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$(3) A^T A x = A^T b$$

$$\begin{pmatrix} 9 & 0 \\ 0 & 5 \end{pmatrix} x = \begin{pmatrix} 9 \\ 10 \end{pmatrix}$$

$$x = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

$$六. a_0 = 0 \quad a_1 = 1 \quad a_2 = 1 \quad \beta_0 = -\frac{1}{12} \quad \beta_1 = \frac{9}{12} \quad \beta_2 = \frac{5}{12}$$

$$c_0 = a_0 + a_1 + a_2 = 0 \quad c_1 = a_1 + 2a_2 - (\beta_0 + \beta_1 + \beta_2) = 1 - (-\frac{1}{12} + \frac{9}{12} + \frac{5}{12})$$

$$c_2 = \frac{1}{2}(a_1 + 2^2 a_2) - (\beta_1 + 2\beta_2) = \frac{5}{2}(\frac{9}{12} + \frac{10}{12}) \quad c_3 = \frac{1}{6}(a_1 + 2^3 a_2) - \frac{1}{2}(\beta_1 + 4\beta_2) = \frac{7}{6} - \frac{1}{2}(\frac{9}{12} + \frac{5}{3})$$

$$c_4 = \frac{1}{24}(a_1 + 2^4 a_2) - \frac{1}{6}(\beta_1 + 2^3 \beta_2) = \frac{15}{24} - \frac{1}{6}(\frac{9}{12} + \frac{40}{12})$$

$$a = 8 \text{ 时 } c_1 = c_2 = c_3 = 0 \quad c_4 = -\frac{1}{24}$$

$$(2) p(\lambda) = \lambda^2 - \lambda \quad \lambda = 1 \text{ 为单根 } p = 4 > 1 \quad \therefore \text{收敛}$$

$$(3) p(\lambda) - \bar{h} \phi(\lambda) = 0$$

$$\lambda^2 - \frac{H_3^2 \bar{h}}{1 - \frac{5}{12} \bar{h}} \lambda - \frac{(\frac{1}{12} \bar{h})}{1 - \frac{5}{12} \bar{h}} = 0 \quad \left| \frac{H_3^2 \bar{h}}{1 - \frac{5}{12} \bar{h}} \right| < 1 + \frac{\frac{1}{12} \bar{h}}{1 - \frac{5}{12} \bar{h}} < 2 \quad -6 < \bar{h} < 0$$

t.	X	P(X)	-阶均差	=阶均差	=阶均差
	0	1			
	0	1	0	+	
	1	0	-1		2
	1	0	0	1	

$$P(0,0) = P'(0) = 0$$

$$P(1,1) = P'(1) = 0$$

$$P(X) = 1 - X^2 + 2X^2(X-1) = 2X^3 - 3X^2 + 1$$

$$P(1.5) = 1$$